

Customer Churn Prediction Using Machine Learning

Predictive Analytics for Strategic Business Growth

Salah Abdeldaim

DATA SCIENCE COMMITTEE FINAL PROJECT

Project Overview

The Primary Goal

Develop a robust machine learning framework to accurately predict whether a customer will churn or remain with the service provider.

By identifying at-risk customers early, businesses can implement proactive retention strategies.

Strategic Importance

Customer retention is significantly more cost-effective than customer acquisition.

Minimizing churn directly correlates to:

-  Increased Customer Lifetime Value
-  Sustained Market Share
-  Higher Profitability Margins

Dataset Description



User Profile

Features include Age and Gender, providing demographic context to usage patterns.



Engagement

Metrics like Support Calls, Last Interaction, and Contract Length signify customer loyalty.



Financials

Total Spend and Payment Delays act as critical indicators of potential dissatisfaction.



Data Pipeline: The dataset underwent rigorous merging, shuffling, and cleaning to ensure high-quality training data for the models.

Data Preprocessing

- ▼ **Missing Values:** Comprehensive cleaning and imputation.
- </> **One-Hot Encoding:** Converting categorical variables to numerical.
- ✂ **Feature Selection:** Identifying high-impact variables.
- ⚙ **Feature Engineering:** Creating new insights from raw data.
- ↔ **Train/Test Split:** ensuring unbiased model evaluation.
- ↗ **Scaling:** Normalized features for Logistic Regression.

Process Integrity

The preprocessing pipeline transforms raw, noisy data into a structured format suitable for high-performance ML algorithms.

Correlation Analysis

Critical Influencers

Our analysis revealed that specific customer behaviors are highly correlated with churn events:



Support Calls: Highest predictor.



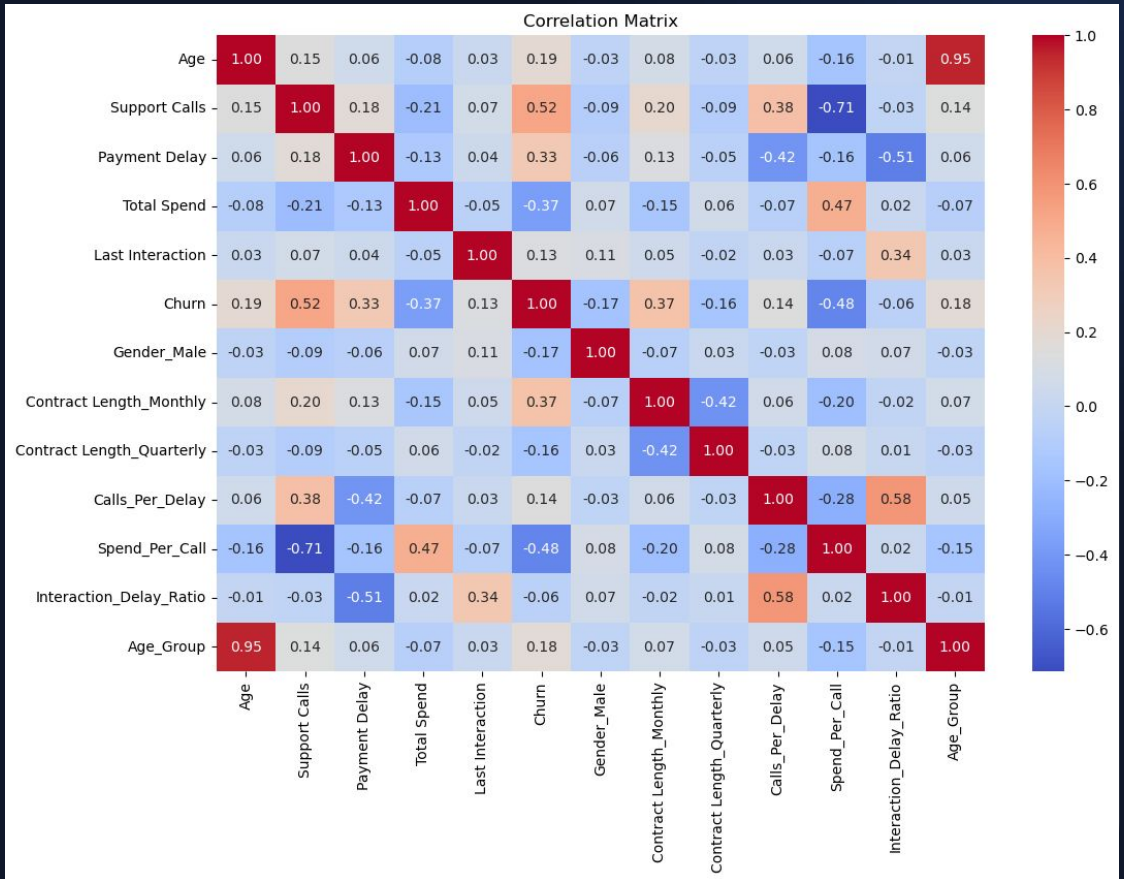
Payment Delay: Significant impact.



Contract Length: Loyalty factor.



Total Spend: Revenue weight.



Models Evaluated

Logistic Regression

Baseline linear model for binary classification.

Decision Tree

Capturing non-linear relationships via splits.

Extra Trees

Randomized ensemble for reduced variance.

XGBoost

High-performance gradient boosting.

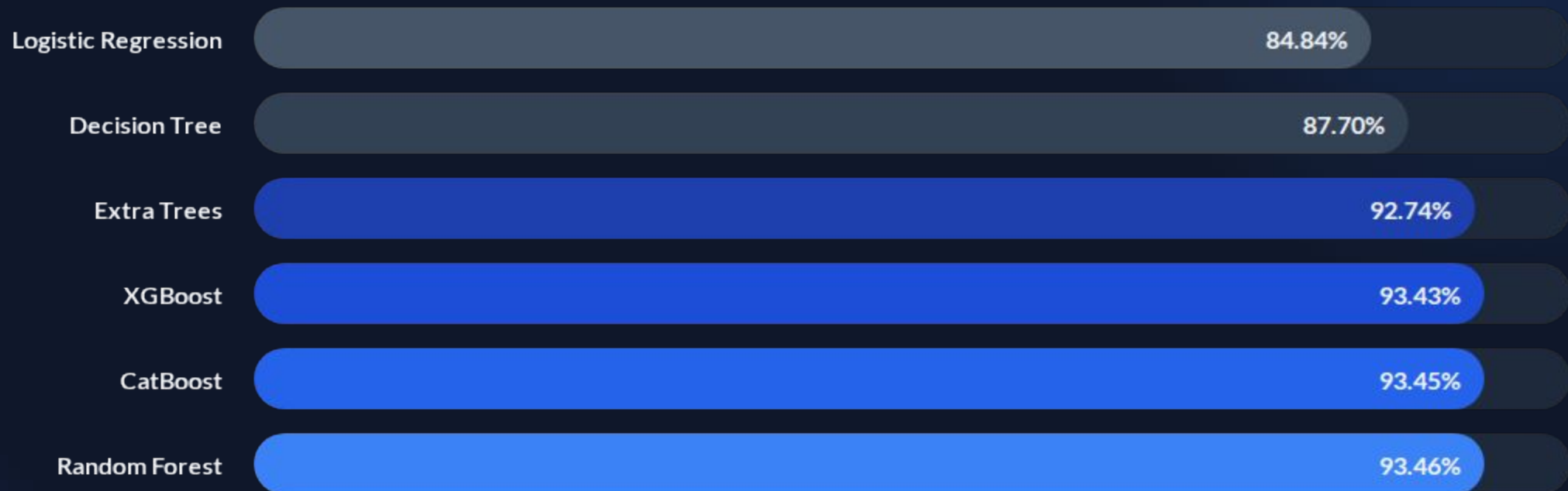
CatBoost

Optimized handling of categorical data.

Random Forest

Robust bagging ensemble approach.

Model Accuracy Comparison



Random Forest achieved the best overall performance with 93.46% accuracy.

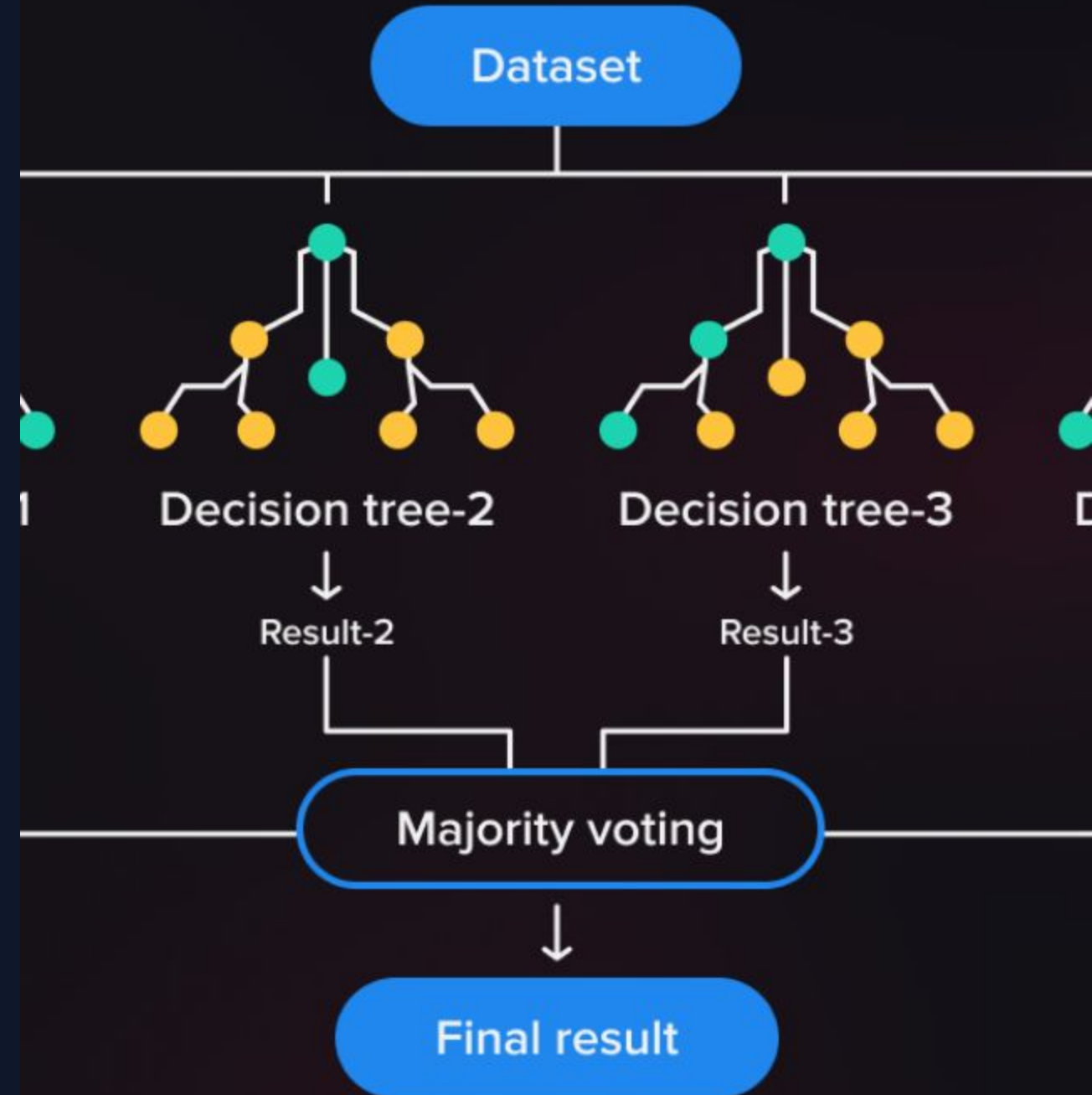
The Champion Model

Random Forest Classifier

Selected for its superior ability to handle high-dimensional data and provide high interpretability through feature importance scores.

Top Features:

- ★ Support Calls
- ★ Total Spend
- ★ Payment Delay
- ★ Contract Length
- ★ Age



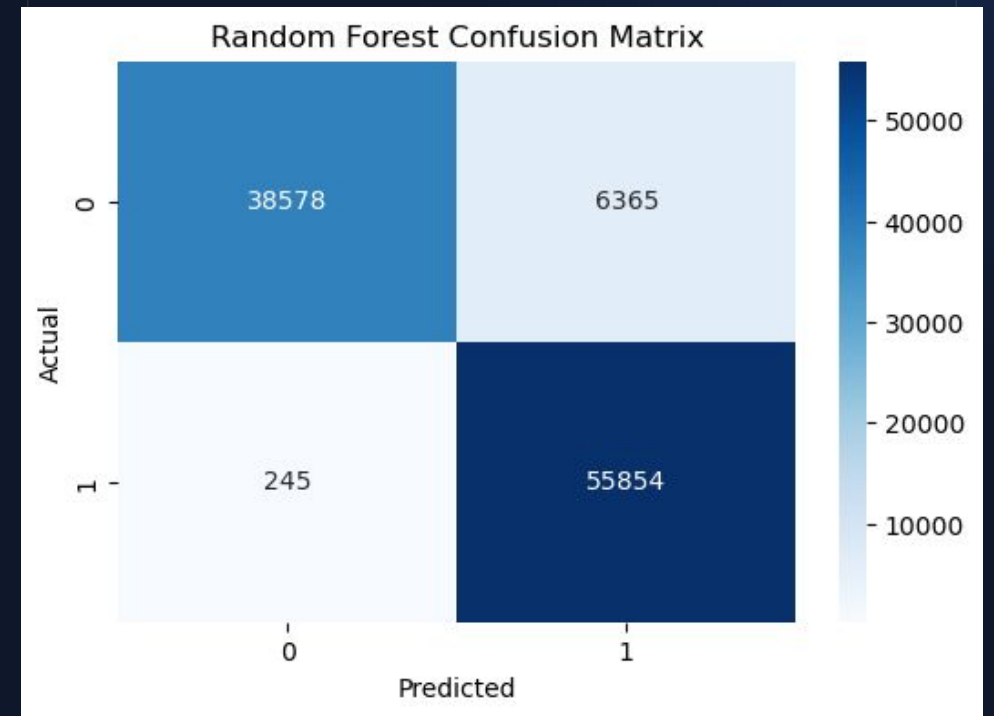
Final Results

93.46%

**Final
Accuracy**

The model demonstrates exceptional precision and recall, ensuring that churners are correctly identified with minimal false positives.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$



Conclusion

- ✓ **Successful Pipeline:** Robust preprocessing and feature engineering.
- 🧠 **Model Benchmarking:** Comparative analysis of multiple state-of-the-art algorithms.
- 🏆 **Optimized Performance:** Random Forest delivered peak predictive power.
- 📁 **Actionable Insights:** Successfully predicting churn with 93.46% reliability.

Thank you! Any Questions?